scientific data

Article in Press

Check for updates

OPEN

DATA DESCRIPTOR

A global 60 m land cover dataset with a fine classification system for 1975 and 1980 derived from Landsat MSS observations

Weitao Ai[1,2,3], Xiao Zhang[2,3] ✉, Chengfei Wang[2,3,4], Fuquan Tang[1], Tingting Zhao[2,3,5], Wenhan Zhang[2,3,4] & Liangyun Liu[2,6]

Long-term land cover datasets are essential for understanding human-environment interactions and global environmental change, yet global land cover information pre-1985 remains scarce because of limited availability of early satellite observations. Here, we present GLC_FCS60, the first global 60 m land cover dataset with a fine classification system for 1975 and 1980, derived from Landsat Multispectral Scanner (MSS) observations. Specifically, an improved comprehensive change detection method with an adaptive thresholding strategy was applied to identify changed and unchanged land-cover pixels. The detected changed areas were then updated using high-confidence temporally stable training samples and locally adaptive classification models on the Google Earth Engine platform. The updated land cover maps were further optimized through post-processing to mitigate potential spurious changes, after which the GLC_FCS60 time-series maps were generated. Finally, using 71,983 globally distributed validation samples, we assessed the accuracy of GLC_FCS60, achieving overall accuracies of 80.13% and 80.08% for the 10 basic classes in 1975 and 1980, respectively, while a dedicated change/no-change validation achieved an overall accuracy of 78.61%. Therefore, GLC_FCS60 fills a critical gap in global land cover observations prior to the mid-1980s and provides a valuable baseline for investigating long-term land cover dynamics from the mid-1970s to the present.

Background & Summary

Land cover data provide essential information for understanding terrestrial surface dynamics and supporting applications such as climate change assessment and ecosystem monitoring[1,2]. Since Landsat's first launch in 1972, remote sensing has enabled consistent and long-term observations of global land surfaces[3,4]. Benefiting from open-access remote sensing archives (e.g., the Landsat series)[5] and the fast advances in cloud platforms including Google Earth Engine (GEE)[6], numerous global land cover datasets have emerged in recent decades[4,7–12]. However, most of these products rely on Landsat Thematic Mapper (TM) and later sensors, which provide higher spectral and radiometric quality and denser temporal coverage than earlier satellite observations[13]. Consequently, existing datasets typically begin in the mid-1980s or later, leaving the pre-TM era—particularly the 1970s and early 1980s—largely unexplored. This gap is particularly important because previous studies have shown that global land-use change accelerated substantially between 1960 and 2004[14], making the early satellite era a critical period for investigating land cover dynamics. Despite its importance, global-scale land cover mapping for this period has remained challenging because of the limitations of early satellite observations, particularly Landsat Multispectral Scanner (MSS) imagery[13]. As a result, our understanding of long-term land cover dynamics remains incomplete.

[1]College of Geomatics, Xi'an University of Science and Technology, Xi'an, 710054, China. [2]International Research Center of Big Data for Sustainable Development Goals, Beijing, 100094, China. [3]Key Laboratory of Digital Earth Science, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, 100094, China. [4]University of Chinese Academy of Sciences, Beijing, 100049, China. [5]School of Geography and Ocean Science, Nanjing University, Nanjing, 210023, China. [6]School of Remote Sensing, Beijing Normal University, Beijing, 100875, China. ✉e-mail: zhangxiao@radi.ac.cn

Dataset	Source	Spatial resolution	Temporal coverage	Access method
Landsat 1–5 MSS	USGS	60 m	1972–1988	https://developers.google.com/earth-engine/datasets/catalog/landsat-mss?hl=zh-cn
Landsat 4–5 TM	USGS	30 m	1982–1988	https://developers.google.com/earth-engine/datasets/catalog/landsat?hl=zh-cn
GLC_FCS30D	Zhang *et al.*	30 m	1985–2022	https://doi.org/10.5281/zenodo.15063683
ASTER GDEM	NASA/METI	30 m	2000-2013	https://doi.org/10.5067/ASTER/ASTGTM.003 (GEE public asset: projects/sat-io/open-datasets/ASTER/GDEM)
SRS_Val	Zhao *et al.*	30 m	2020	https://doi.org/10.5281/zenodo.7846090

Table 1. Third-party datasets used in this study.

Compared with later sensors (such as TM and ETM +), MSS data provide sparser temporal coverage and fewer spectral bands, increasing the difficulty of accurate land cover mapping and change monitoring. Previous studies have successfully applied MSS imagery to regional land cover mapping, often relying on limited observations and visual interpretation[15,16]. However, the substantial time, labor, and financial resources required by these approaches[17] pose a major challenge for global-scale applications, where spatial consistency and classification accuracy should be maintained. Although the feasibility of using MSS data for local-scale mapping has been demonstrated, an automated framework capable of supporting consistent global-scale land cover mapping from MSS observations is still lacking. Therefore, there is a clear need to fully exploit MSS imagery, develop mapping approaches tailored to the characteristics of early satellite observations, and generate globally consistent land cover products for the pre-1985 period.

For monitoring global land cover changes, two main strategies are commonly adopted: "independent multi-temporal classification" and "change-detection-then-classification"[13]. The former classifies each epoch independently, often suffering from error accumulation and temporal inconsistency caused by classification noise and sensor differences[18,19]. In contrast, the latter identifies changed pixels and subsequently updates their labels using a reference map, while unchanged pixels directly inherit their labels from the reference map, thereby improving temporal consistency and reducing spurious changes[18,20,21]. Within this strategy, the selection of change detection algorithm is strongly influenced by the availability of satellite observations. When dense time-series observations are available (e.g., post-2000 Landsat or Sentinel archives), trajectory-fitting approaches like the Continuous Change Detection and Classification (CCDC) algorithm can effectively capture both gradual and abrupt changes by modeling temporal trajectories[22]. However, for the early satellite era, when only sparse observations are available, bi-temporal change detection methods are generally more appropriate. Given the limited and irregular temporal coverage of MSS imagery in the 1970s and early 1980s, a bi-temporal approach represents a practical and effective solution. Therefore, in this study, we adopt an improved bi-temporal Comprehensive Change Detection Method (CCDM)[13,23] to identify changed pixels between 1975/1980 and 1985, leveraging the high-quality 1985 land cover map from GLC_FCS30D as a reference baseline.

To fill this gap, we generated GLC_FCS60, the first global land cover dataset with a fine classification system at 60 m resolution for 1975 and 1980, derived primarily from Landsat MSS imagery. The study comprised two main components: (1) constructing GLC_FCS60 using a change-detection-then-classification strategy based on the improved CCDM[13]; and (2) conducting a comprehensive accuracy assessment of the products.

Methods

Remote sensing datasets. To generate GLC_FCS60, we used multi-source remote sensing imagery and ancillary data. The basic information, sources, and access methods of these datasets are summarized in Table 1, while detailed descriptions of the individual datasets are provided below.

Landsat MSS archives constituted one of the primary medium-resolution remote sensing data sources from the 1970s to the early 1980s. First launched in 1972 and providing imagery at 60 m resolution, MSS served as a critical foundation for early land cover mapping efforts. Landsat MSS imagery includes four spectral bands: one green band, one red band, and a pair of near-infrared (NIR) bands. In this study, Landsat MSS imagery was used as the core dataset for generating GLC_FCS60. Given the sparse temporal coverage and frequent data gaps in MSS imagery, all available observations within a ± 3-year window centered on each base year were used to generate image composites. This strategy improves spatial coverage and reduces the influence of noise and anomalous observations. Figure 1 shows the global distribution of MSS false-color composites for 1975, 1980, and 1985 generated using the GEE platform. Despite the use of multi-year compositing, some regions still lacked valid MSS observations during these periods. These areas accounted for 19.88% and 25.29% of the global mapped area in 1975 and 1980, respectively. In regions where MSS imagery was unavailable in 1985 (e.g., large parts of western China), Landsat TM imagery was used as a substitute reference dataset for change detection against the target-year composites. To maintain consistency with MSS imagery, the green, red, and NIR bands were selected from the TM imagery, and its 30 m spatial resolution was subsequently down-sampled to 60 m. All MSS and TM imagery were obtained through the GEE platform.

The global land cover product GLC_FCS30D[4] was used as an important auxiliary data source. This 30-m product provides global land cover maps from 1985 to 2022 with a fine classification system and high overall accuracy. Consistent with the classification system of GLC_FCS30D, the dataset serves two key roles in the mapping process. First, it is used to identify stable pixels that remain unchanged over extended periods, from which high-confidence training samples are automatically generated. Second, it provides prior information for constraining the change detection process by guiding the selection of adaptive thresholds, thereby ensuring that the magnitude of detected changes in earlier periods remains consistent with long-term land cover dynamics.

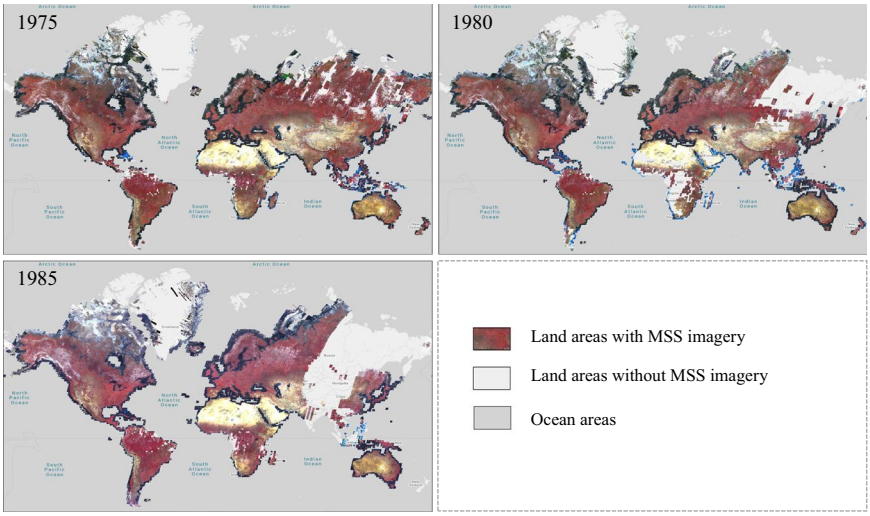


Fig. 1 Global distribution of Landsat MSS false-color composites for 1975, 1980, and 1985.

In this study, the post-processed GLC_FCS30D product[24] was used and can be freely accessed via the Zenodo repository (https://doi.org/10.5281/zenodo.15063683)[25].

The inclusion of topographic variables is important because land cover distribution is strongly influenced by topographic conditions, particularly in mountainous regions where elevation and slope affect vegetation types and land use patterns. In this study, ASTER Global Digital Elevation Model (GDEM)[26] data available through the GEE platform were used to derive topographic variables, including elevation, slope, and aspect.

The SRS_Val global validation dataset developed by Zhao *et al.*[27] was used for accuracy assessment. It was generated using a stratified random sampling strategy that accounted for landscape heterogeneity and population density, providing representative global spatial coverage. The dataset contains reference samples for 2020, which were further screened and interpreted in this study to derive reference samples for evaluating the 1975 and 1980 GLC_FCS60 products. The SRS_Val dataset can be freely accessed via the Zenodo repository (https://doi.org/10.5281/zenodo.7846090)[28].

Methodological framework. Figure 2 illustrates the overall workflow for generating GLC_FCS60 from Landsat MSS imagery. The framework consists of four main stages: (1) image compositing and normalization; (2) application of the improved bi-temporal CCDM to identify changed and unchanged pixels; (3) local adaptive classification of changed pixels; and (4) post-classification processing to generate the consistent land cover maps. For areas without valid MSS observations for a target year, where bi-temporal change detection could not be performed, the corresponding land-cover labels were directly inherited from GLC_FCS30D_1985 under a no-change assumption.

Image compositing and normalization. Multi-year image compositing was performed to generate cloud-free images before bi-temporal change detection. Landsat MSS observations during the early satellite era were sparse and spatially incomplete, and were often contaminated by clouds and shadows, making single-date imagery insufficient for large-area mapping. The valid observations were therefore composited within a predefined multi-year temporal window to generate a high-quality image for each target epoch. Following previous studies[29,30], composites were generated separately for the growing and non-growing seasons to support accurate change detection. In this study, all available Landsat MSS and TM scenes within the compositing window were first masked to remove clouds, shadows, and other invalid observations. The median-value compositing[29,30] approach was then applied separately to the growing- and non-growing-season observations to generate the corresponding composite images. It is noteworthy that the growing season was defined as April–October (Northern Hemisphere) and October–April (Southern Hemisphere), while the remaining months were assigned to the non-growing season.

As subsequent change detection uses 1985 as the reference baseline, Landsat TM imagery was incorporated in regions where valid MSS observations were unavailable. Although MSS and TM are both Landsat sensors, they differ in spectral response and spatial resolution. Furthermore, radiometric inconsistencies may exist among MSS observations acquired by different Landsat missions under varying acquisition conditions. Therefore, we applied the Iteratively Reweighted Multivariate Alteration Detection (IR-MAD)[31] algorithm combined with geometric distance-based orthogonal regression[32] to normalize the 1975 and 1980 composites against the 1985 reference composites. Specifically, IR-MAD was first used to automatically identify pseudo-invariant pixels between the target and reference images. These invariant pixels were then used to construct orthogonal regression models, which transformed the target imagery to be radiometrically consistent with the reference imagery. In addition, when both the target-year and 1985 composites were derived from MSS imagery, all four MSS spectral bands were retained. When 1985 TM composites were used instead, only the green, red, and NIR-1

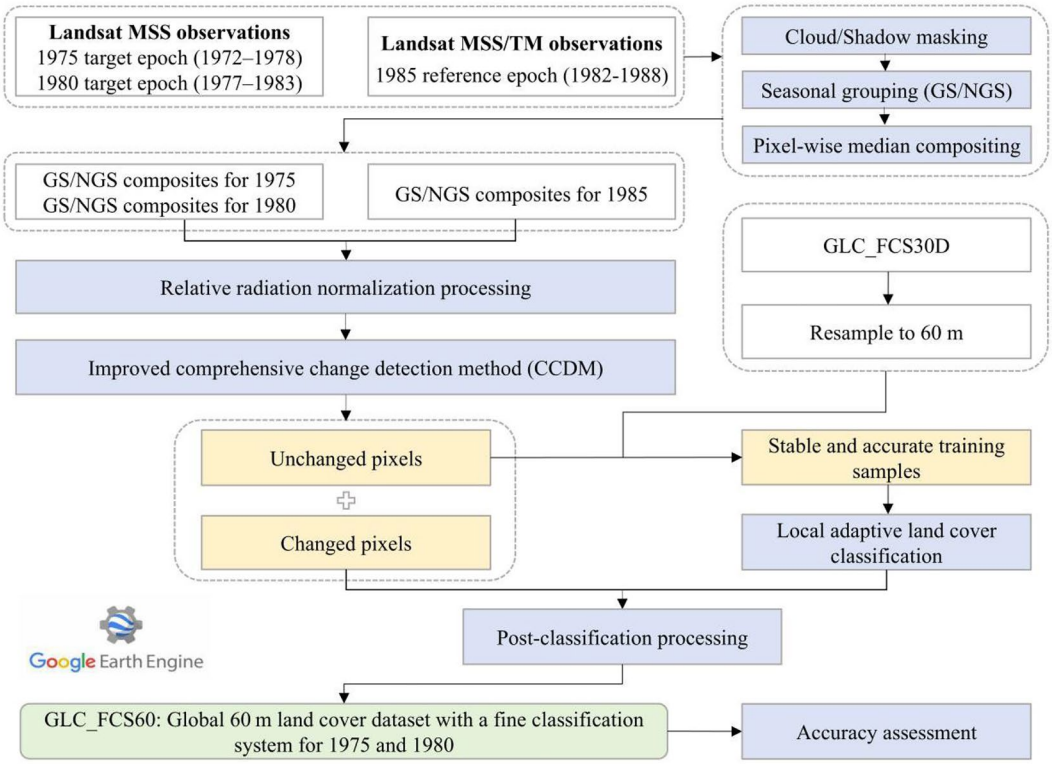


Fig. 2 Workflow for generating GLC_FCS60, where GS and NGS denote growing season and non-growing season, respectively.

bands of MSS imagery were retained, whereas the NIR-2 band was excluded to maintain spectral consistency with TM observations.

Improved Comprehensive Change Detection Method. To obtain reliable land-cover change information under sparse MSS observations, an improved CCDM[13] was applied by comparing the composited and normalized imagery from the target years (1975 and 1980) with the 1985 reference imagery. The method consists of three main components: (1) the Multi-Index Integrated Change Analysis (MIICA) model, (2) the Zone change detection model, and (3) an integration of MIICA and Zone outputs via an adaptive thresholding strategy.

The original CCDM integrates four spectral change indices: the differenced Normalized Difference Vegetation Index (dNDVI), differenced Normalized Burn Ratio (dNBR), Change Vector (CV), and Relative Change Vector Maximum (RCVMAX), to characterize land cover changes for both growing and non-growing seasons. Notably, since the MSS sensor lacks shortwave infrared (SWIR) bands, and some previous studies have shown that the Green Normalized Difference Vegetation Index (GNDVI) is sensitive to vegetation conditions and changes[33,34], dGNDVI was used in place of dNBR for capturing vegetation-related changes. The four indices were calculated using Eqs. (1–4).

$$dNDVI_t = NDVI_{ref,t} - NDVI_{tar,t}, \quad NDVI = (\rho_{NIR} - \rho_{Red})/(\rho_{NIR} + \rho_{Red}) \tag{1}$$

$$dGNDVI_t = GNDVI_{ref,t} - GNDVI_{tar,t}, \quad GNDVI = (\rho_{NIR} - \rho_{Green})/(\rho_{NIR} + \rho_{Green}) \tag{2}$$

$$CV_t = \sum_i (\rho_{ref,t,i} - \rho_{tar,t,i})^2 \tag{3}$$

$$RCVMAX_t = \sum_i [(\rho_{ref,t,i} - \rho_{tar,t,i})/\max(\rho_{ref,t,i}, \rho_{tar,t,i})^2] \tag{4}$$

where *ref* and *tar* denote the reference year (1985) and the target year (1975 or 1980), respectively; ρ_{Green}, ρ_{Red} and ρ_{NIR} denote surface reflectance values; the subscript t indicates the season (growing or non-growing); and i is the serial number of corresponding spectral bands. The magnitude of each index reflects the intensity of land-cover change, with larger magnitudes indicating a higher likelihood of change. However, considering only change intensity may overlook valuable directional information. In particular, both dNDVI and dGNDVI can reveal the direction of land-cover change. A positive dNDVI value suggests a decrease in biomass, potentially indicating processes such as deforestation during this period. Building on previous work[13,23], this directional

information was further incorporated into the MIICA model to distinguish two change types: biomass increase (BI) and biomass decrease (BD), as defined in Eqs. (5–6).

$$BI_{t-MIICA} = (CV > \mu_{CV}) \bigcap (RCVMAX > \mu_{RCVMAX} + n_1 \times \sigma_{RCVMAX}) \bigcap (dNDVI < \mu_{dNDVI} - n_2 \times \sigma_{dNDVI}) \quad (5)$$

$$BD_{t-MIICA} = (CV > \mu_{CV}) \bigcap (RCVMAX > \mu_{RCVMAX} + n_1 \times \sigma_{RCVMAX}) \bigcap (dNDVI > \mu_{dNDVI} + n_2 \times \sigma_{dNDVI}) \quad (6)$$

where μ and σ denote the mean and standard deviation of the corresponding index, respectively; and n_1 and n_2 are parameters controlling the sensitivity of change detection in the MIICA model.

The change information extracted by MIICA is not fully comprehensive. Although both dGNDVI and dNDVI can indicate the magnitude and direction of land-cover change, dGNDVI is not fully utilized in the change detection process. Therefore, we introduced the Zone model, which jointly utilizes dGNDVI and dNDVI to compensate for potential omission errors in MIICA[13,23]. Specifically, using these two indices, biomass changes are grouped into four categories: weak increase (B_+), weak decrease ($B_$), strong increase (B_{++}), and strong decrease ($B__$). The corresponding value ranges are defined in Eq. (7).

$$\begin{aligned} B_+ &\in (\mu - n_3 \times \sigma, \mu] \\ B_ &\in (\mu, (\mu + n_3 \times \sigma) \\ B_{++} &\in (-\infty, \mu - n_3 \times \sigma] \\ B__ &\in [\mu + n_3 \times \sigma, (+\infty) \end{aligned} \quad (7)$$

where n_3 is the parameter controlling the sensitivity of change detection in the Zone model. Furthermore, if the two indices exhibit consistent directions and both indicate strong increase or strong decrease, the pixel is classified as BI or BD, respectively, according to Eqs. (8–9).

$$BI_{t-Zone} = B_{++}(dNDVI) \bigcap B_{++}(dGNDVI) \quad (8)$$

$$BD_{t-Zone} = B__(dNDVI) \bigcap B__(dGNDVI) \quad (9)$$

As a complement to MIICA, the Zone model was integrated with it. The union of change outputs from the two models served as the change detection result of the improved CCDM. Furthermore, to ensure the reliability of detected changes, only pixels identified as changed in both growing-season and non-growing-season analyses were retained to form the preliminary change mask, namely the intersection of the two seasonal change maps.

It is worth noting that both MIICA and Zone are parameter-sensitive methods, and their associated parameters (n_1, n_2, and n_3 in Eqs. (5–7)) directly influence detection reliability. Conventional approaches rely on fixed empirical thresholds[23], which are inherently subjective and may not adapt well to regional variations. Therefore, in this study we proposed using the GLC_FCS30D product as a priori knowledge to guide parameter determination. Specifically, as land cover change is closely associated with human activities, short-term changes are generally expected to be smaller than long-term ones. Based on this rationale, land cover changes from 1975/1980 to 1985 (Ch_pre) were assumed not to exceed those observed during 1985–2022 (Ch_late)[13]. The parameters in MIICA and Zone were then adaptively adjusted for each 5° × 5° tile such that Ch_pre was as close as possible to, but did not exceed, the corresponding Ch_late derived from GLC_FCS30D. The optimal values of n_1, n_2, and n_3 were selected for each tile, thereby generating final change masks that preserved directional information (BI/BD) for subsequent classification and analysis.

Local adaptive classification for changed pixels. Land cover mapping accuracy relies heavily on the quality of the training samples used[35,36]. While visual interpretation can provide highly accurate samples, this manual approach involves a heavy workload and long processing cycles, limiting its applicability to global-scale mapping[37]. To overcome this limitation, a globally distributed pool of high-confidence training samples was automatically generated using existing land cover products. Recent studies have demonstrated that spatiotemporally stable regions can provide reliable training samples[38,39]. Following this rationale, stable training samples were generated by intersecting two sets of pixels: (1) pixels identified as unchanged by the improved CCDM between the target year (1975/1980) and 1985, and (2) pixels with consistent land cover labels in the 1985–2000 GLC_FCS30D product (resampled to 60 m to match MSS imagery). The land cover labels of these unchanged pixels were directly derived from the GLC_FCS30D map in 1985. To ensure representativeness and reduce sample imbalance, the global land area was split into 961 geographical tiles of 5° × 5°, following the zoning scheme adopted in GLC_FCS30D[4]. Within each tile, training samples were allocated using an area-proportional strategy, with minimum and maximum sample sizes of 600 and 8,000 samples per land cover type, respectively, following previous studies[40–42].

For the classification of changed pixels, a local adaptive modeling strategy was adopted, whereby a separate random forest classifier was constructed for each 5° × 5° tile. This is because local adaptive models can more effectively capture regional complexity and heterogeneity and improve classification accuracy[39,43]. Additionally, random forest is an ensemble classifier that can effectively alleviate overfitting issues and exhibit low sensitivity to noisy training data[38,44,45]. As a result, a total of 21 multi-source features were fed into the random forest classification model. These features included 18 multi-temporal spectral variables derived from Landsat MSS composites, consisting of four spectral bands (green, red, NIR-1, and NIR-2) and two vegetation indices (NDVI

and GNDVI) calculated from the growing-season, non-growing-season, and annual-median composites. The remaining three topographic features—elevation, slope, and aspect—were obtained from the ASTER GDEM. Random forest classification was executed on the GEE platform using the ee.Classifier.smileRandomForest() function, with 100 decision trees[46]. The final land cover maps for 1975 and 1980 were generated by merging unchanged pixels, whose labels were directly inherited from the 1985 GLC_FCS30D map, with the newly classified changed pixels. For areas lacking MSS imagery coverage in 1975 and 1980, land cover types were assumed to remain unchanged from 1985.

Post-classification processing. To reduce spurious changes and improve the temporal consistency of the 1975 and 1980 land cover maps, two post-processing steps were applied. First, many previous studies treat impervious surface expansion as an irreversible process[43,47]; accordingly, we also imposed an irreversibility constraint on impervious surfaces in this study. Transitions from non-impervious surfaces in 1975 or 1980 to impervious surfaces in 1985 were retained. In contrast, transitions from impervious surfaces in 1975 or 1980 to non-impervious surfaces in 1985 were treated as unrealistic and constrained during post-processing, with these cases reassigned to non-impervious surfaces. The same constraint was applied between 1975 and 1980 to ensure that impervious surface extent in 1975 did not exceed that in 1980.

Second, a modified version of the SKILL model (mSKILL)[13,48] was adopted to correct ecologically implausible land cover transitions by incorporating biomass change information from the improved CCDM outputs. Here, dNDVI was used as a proxy for the magnitude and direction of biomass change. The mSKILL model was designed to address two main types of inconsistencies: (1) Transitions that implied biomass increase (grassland to shrubland, shrubland to forest) but where the change detection indicated a decrease, and conversely, transitions implying biomass decrease (forest to shrubland, forest to grassland, shrubland to grassland) where the change detection indicated an increase. For such directional inconsistencies, the land cover type in 1975/1980 was reverted to the 1985 class (i.e., treated as unchanged). (2) Conversions from forest to grassland versus forest to shrubland, which required distinguishing the severity of biomass loss. For instance, if a pixel was classified as forest in 1985 but as grassland or shrubland in 1975/1980, the expected change direction was biomass decrease. The severity of biomass loss was then determined using the mean (μ) and standard deviation (σ) of dNDVI: a dNDVI value exceeding $\mu + 2\sigma$ indicated high-severity loss, and the pixel was labeled as grassland; a dNDVI value between $\mu + \sigma$ and $\mu + 2\sigma$ indicated moderate-severity loss, and the pixel was labeled as shrubland. This procedure effectively reduced spurious changes and enhanced the ecological plausibility of land cover transitions for the 1975 and 1980 maps.

Data Records

The 60 m global land cover datasets for 1975 and 1980 can be freely accessed through the Zenodo platform (https://doi.org/10.5281/zenodo.20688228)[49]. To facilitate efficient storage and data access, the data are organized into 961 geographical tiles of 5° × 5°. Each tile is provided as a GeoTIFF file containing two bands, where Band 1 and Band 2 correspond to land cover maps for 1975 and 1980, respectively. The tiles are named "GLC_FCS60_19751980_E***N##", where "***" and "##" denote the longitude and latitude coordinates of the upper-left corner of each tile (e.g., E120N30). The coordinate reference system (CRS) is WGS84 (EPSG:4326) in geographic coordinates. Land cover types are encoded using integer values consistent with the GLC_FCS30D classification system[4], including 10 basic classes, 17 level-1 classes, and 35 fine classes. Figure 3 presents the global 60 m land cover map for 1975 based on the fine classification system. Details of the classification system are provided in Table S1 of the Supplementary information.

Technical Validation

Global validation dataset collection. Constructing a reliable validation dataset for land cover products in the early satellite era is particularly challenging, primarily due to the scarcity of contemporaneous reference data and the limited availability of satellite observations.

To evaluate the accuracy of GLC_FCS60, the SRS_Val[27] reference samples were further screened and interpreted to derive reliable reference labels for 1975 and 1980. Considering the differences in spatial resolution (30 m versus 60 m) and temporal coverage between SRS_Val and GLC_FCS60, three screening and interpretation steps were implemented to improve sample reliability. First, spatial homogeneity was assessed using the GLC_FCS30D products[4]. For each candidate point, a 3 × 3 pixel window centered on the point was examined, and points with inconsistent land-cover labels within the window were discarded. Second, temporal consistency was evaluated using the multi-temporal GLC_FCS30D products by retaining only points for which the land-cover labels of the corresponding pixels remained unchanged throughout 1985–2022. These two steps yielded 71,983 spatiotemporally homogeneous candidate samples. Finally, potential change areas were identified by comparing the 1975, 1980, and 1985 land-cover maps, and all candidate samples located within these areas were visually interpreted to assign reliable reference labels for 1975 and 1980. Additionally, 10% of the samples located in stable areas were randomly selected and manually checked to verify that no land-cover changes had occurred.

To ensure the confidence of each validation point, we utilized multiple-sourced auxiliary datasets to support the visual interpretation of the relevant sample points. These datasets included: (1) Landsat MSS false-color composites for 1975, 1980, and 1985; (2) NDVI time-series profiles derived from these composites; (3) ASTER GDEM-derived topographic variables; (4) historical climate data; (5) the geographic distribution characteristics of different land-cover subclasses; and (6) the prior land-cover labels associated with the validation samples. Importantly, the prior land-cover labels provided important contextual information for evaluating the plausibility of interpreted changes. The climatic and geographic distribution characteristics of different subclasses

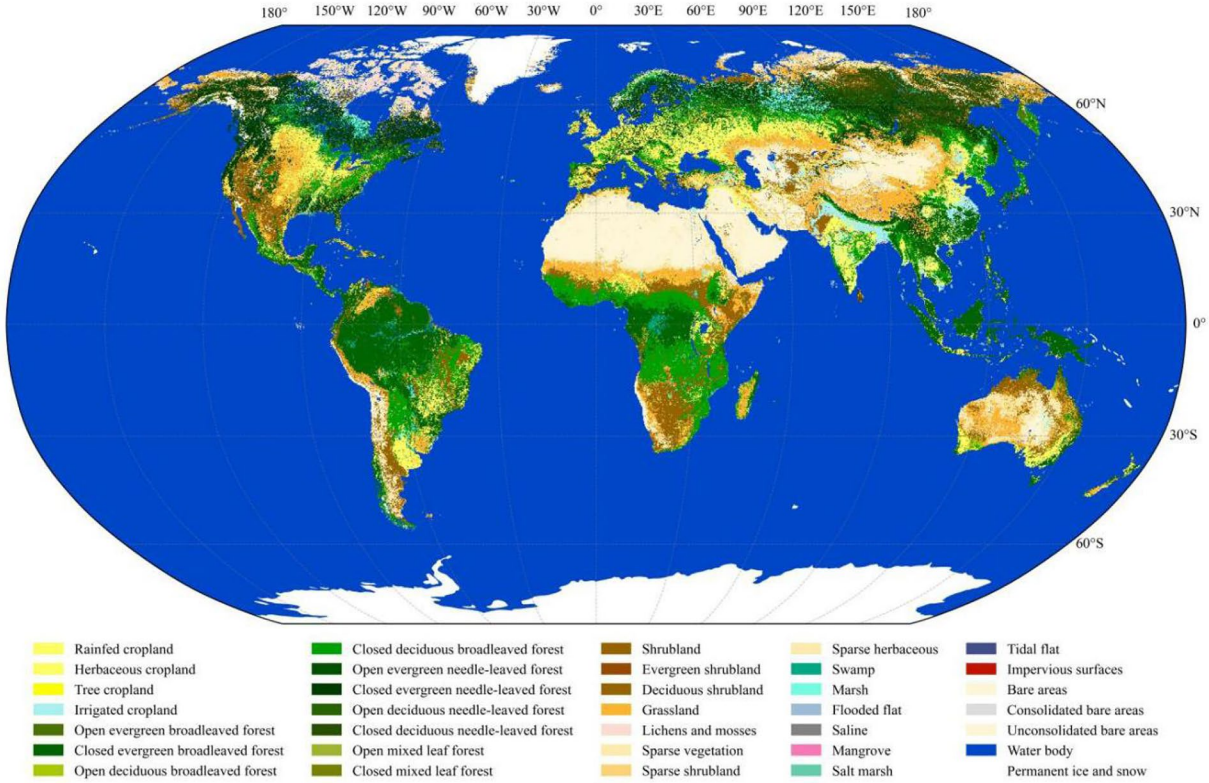


Fig. 3 Overview of the GLC_FCS60 land cover map for 1975.

were also considered to assess whether an interpreted transition was environmentally plausible. For example, when distinguishing between coniferous and broadleaf forests, we considered multiple lines of evidence: coniferous forests are generally associated with colder climates and higher-latitude or high-elevation environments (e.g., boreal and alpine zones), whereas broadleaf forests are more commonly distributed in warmer and more humid climates within temperate, subtropical, and tropical regions. This climatic and geographic context was further integrated with seasonal NDVI trajectories—coniferous forests exhibit relatively stable NDVI throughout the year, while broadleaf forests show clear seasonal greening and senescence patterns—and cross-checked against topographic variables to reach a confident interpretation. When a transition was considered, it was not determined solely from the MSS spectral response, but was evaluated together with the regional climatic and geographic conditions, temporal observations, and the plausibility of the land-cover transition. In the absence of supporting evidence, such a transition would be considered less plausible. The visual interpretation process was conducted with the aid of the interpretation platform developed by Zhao *et al.*[50]. Overall, this procedure produced a validation dataset with improved reference-label reliability, whose spatial distribution broadly represented global land-cover patterns in 1975 (Fig. 4).

To quantify the ability of GLC_FCS60 to detect land-cover changes, a dedicated change/no-change accuracy assessment was conducted. Potential change and no-change strata were first delineated by comparing the land-cover maps for 1975, 1980, and 1985. Candidate samples from the potential change strata were extracted from the validation dataset constructed in the previous step, while samples from the no-change strata were randomly selected near potential change areas to achieve an approximate 3:1 no-change-to-change sample ratio to mitigate class imbalance. All selected samples were then visually interpreted to determine whether actual land-cover change had occurred. This procedure yielded a change/no-change validation dataset comprising 1,664 samples, of which 411 were changed and 1,253 were unchanged (Fig. 5).

Accuracy assessment using the global validation dataset. Error matrices are widely used for assessing classification accuracy[51]. The associated accuracy metrics include producer's accuracy (PA), user's accuracy (UA), overall accuracy (OA) (Eq. (10)), and the Kappa coefficient.

$$\begin{aligned} P.A._{\cdot k} &= \frac{Nkk}{\sum N.k} \\ U.A._{\cdot k} &= \frac{Nkk}{\sum Nk.} \\ O.A. &= \frac{\sum_{k=1}^{m} Nkk}{N} \end{aligned} \tag{10}$$

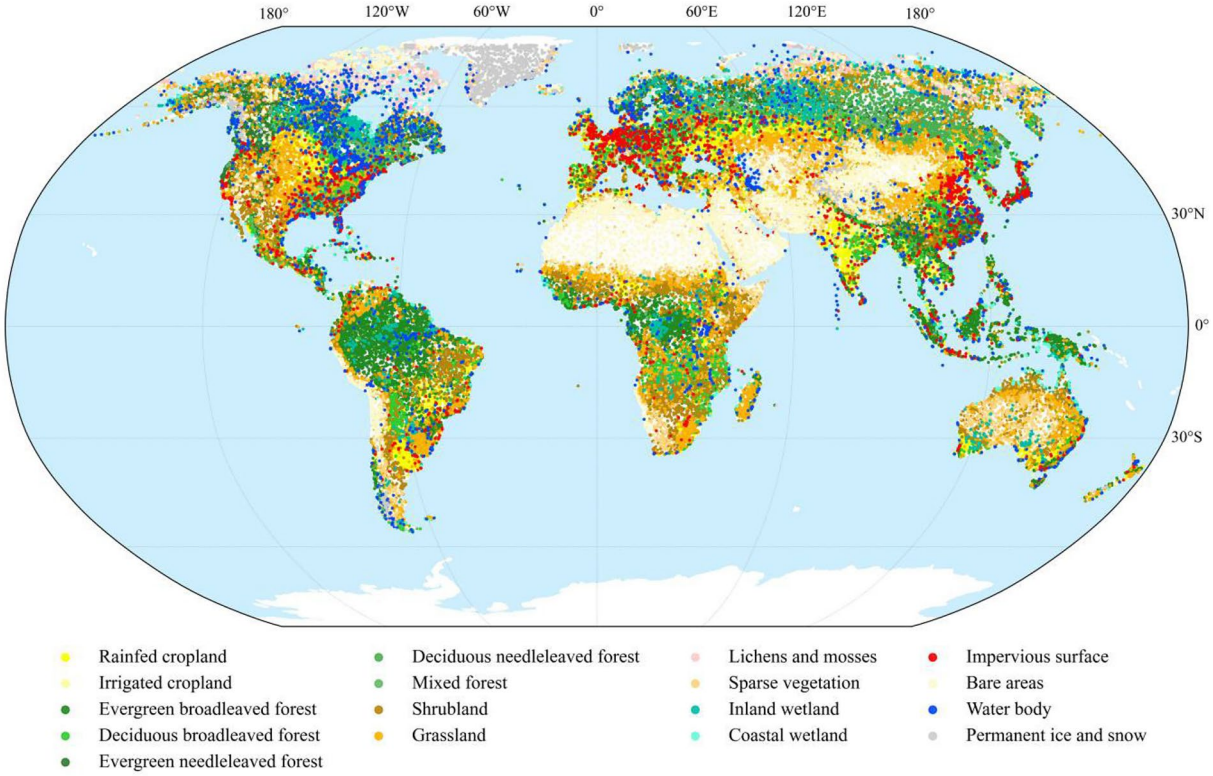


Fig. 4 Global spatial distribution of validation samples covering 17 level-1 land cover classes in 1975.

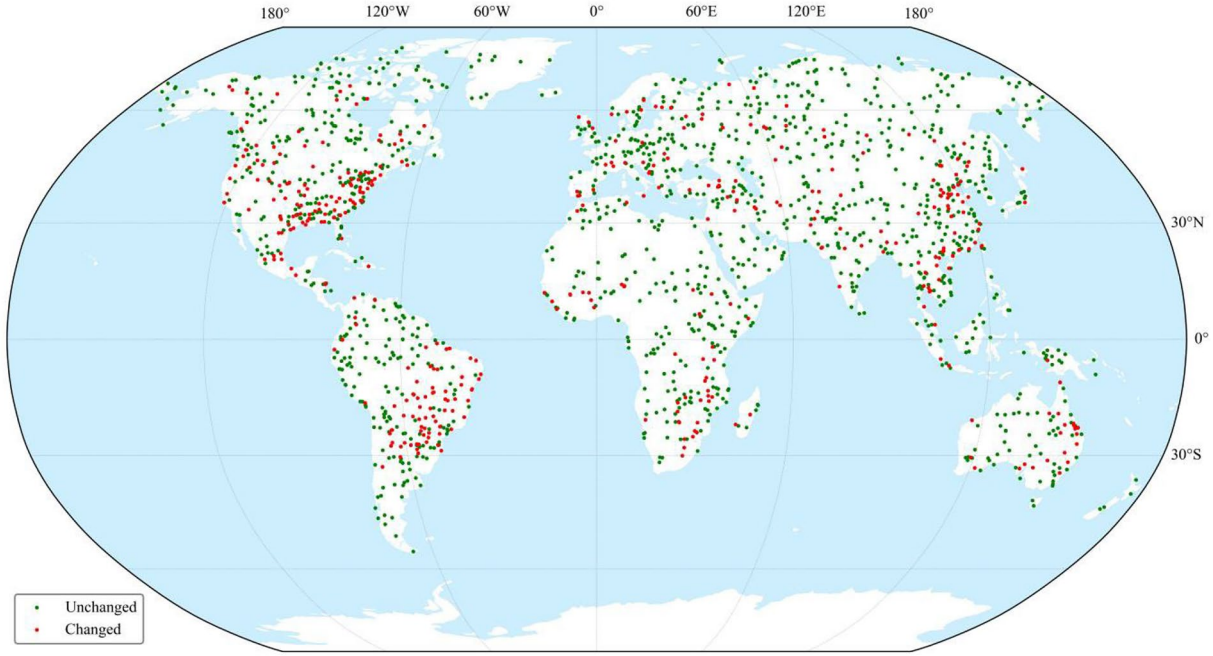


Fig. 5 Global distribution of changed and unchanged validation samples for 1975–1985.

Where *Nkk* is the number of samples correctly classified into class *k*; *N* is the total number of samples; ΣN. *k* is the actual number of samples belonging to class *k*; ΣNk. is the number of samples classified into class *k*; and *m* is the number of land cover types.

Using the 71,983 validation samples, we evaluated the accuracy of GLC_FCS60 for 1975 at two classification levels: 10 basic classes and 17 level-1 classes. The error matrix for the 10 basic classes is presented in Table 2. For the 10 basic classes, the OA reached 80.13%, with a Kappa coefficient of 0.76. Both PA and UA exceeded 80% for

Reference	Map	O.A. = 80.13% Kappa = 0.76										
	CRP	FST	GRS	SHR	WET	WTR	TUD	IMP	BAL	PSI	Total	P.A. (%)
CRP	10778	596	805	250	48	2	0	14	60	0	12553	85.86
FST	261	22361	239	350	456	15	2	14	31	0	23729	94.23
GRS	669	878	4278	873	151	6	50	7	834	1	7747	55.22
SHR	307	1348	665	3577	120	5	9	4	365	1	6401	55.88
WET	44	289	112	111	2432	168	27	6	143	1	3333	72.97
WTR	40	76	12	18	262	2083	9	0	21	1	2522	82.59
TUD	4	106	147	121	13	17	1895	2	467	5	2777	68.24
IMP	312	93	53	35	7	4	0	1870	25	0	2399	77.95
BAL	61	31	890	516	16	17	474	1	7324	46	9376	78.11
PSI	1	3	25	0	0	12	1	0	21	1083	1146	94.50
Total	12477	25781	7226	5851	3505	2329	2467	1918	9291	1138		
U.A. (%)	86.38	86.73	59.20	61.13	69.39	89.44	76.81	97.50	78.83	95.17		

Table 2. Error matrix derived from the 10 basic land cover classes.

Reference	Map	O.A. = 71.60% Kappa = 0.68																	
	RCP	ICP	EBF	DBF	ENF	DNF	MFT	SHR	GRS	LMS	SVG	IWL	CWL	IMP	BAL	WTR	PSI	Total	P.A. (%)
RCP	8287	768	200	305	56	7	0	242	785	0	41	30	4	10	17	1	0	10753	77.07
ICP	356	1367	18	9	1	0	0	8	20	0	2	11	3	4	0	1	0	1800	75.94
EBF	86	25	7193	528	178	75	255	112	50	0	3	165	15	5	0	0	0	8690	82.77
DBF	115	7	501	5214	332	141	203	117	116	0	2	109	1	8	1	3	0	6870	75.90
ENF	19	3	129	133	4410	199	142	53	23	2	11	123	0	1	2	11	0	5261	83.82
DNF	6	0	0	113	214	1369	51	62	48	0	10	27	0	0	2	1	0	1903	71.94
MFT	0	0	21	88	109	280	485	6	2	0	0	14	0	0	0	0	0	1005	48.26
SHR	295	12	268	666	286	123	5	3577	665	9	312	116	4	4	53	5	1	6401	55.88
GRS	626	43	145	450	166	116	1	873	4278	50	676	146	5	7	158	6	1	7747	55.22
LMS	2	2	1	20	29	55	1	121	147	1895	376	13	0	2	91	17	5	2777	68.24
SVG	40	3	5	6	6	2	0	345	553	31	1936	5	0	1	577	4	5	3519	55.02
IWL	30	12	78	51	115	33	1	105	110	25	106	1660	159	4	24	138	1	2652	62.59
CWL	0	2	6	3	1	1	0	6	2	2	9	51	562	2	4	30	0	681	82.53
IMP	235	77	31	22	35	4	1	35	53	0	12	6	1	1870	13	4	0	2399	77.95
BAL	13	5	5	4	2	1	0	171	337	443	538	11	0	0	4273	13	41	5857	72.96
WTR	26	14	17	17	34	8	0	18	12	9	13	204	58	0	8	2083	1	2522	82.59
PSI	0	1	0	2	1	0	0	0	25	1	12	0	0	0	9	12	1083	1146	94.50
Total	10136	2341	8618	7631	5975	2414	1145	5851	7226	2467	4059	2691	812	1918	5232	2329	1138		
U.A. (%)	81.76	58.39	83.46	68.33	73.81	56.71	42.36	61.13	59.20	76.81	47.70	61.69	69.21	97.50	81.67	89.44	95.17		

Table 3. Error matrix derived from the 17 level-1 land cover classes.

cropland, forest, water body, and permanent snow/ice. Grassland and shrubland exhibited moderate accuracies (PA ≈ 55%, UA ≈ 59–61%), primarily because of spectral confusion between these classes and bare land. Impervious surface achieved a high UA of 97.50% but a relatively lower PA of 77.95%, with approximately 13% of reference impervious surface samples misclassified as cropland. This omission error is primarily attributable to the majority-rule aggregation of the 30 m reference product to 60 m resolution, which can merge small rural impervious surface patches into surrounding cropland.

The error matrix for the 17 level-1 classes is presented in Table 3. For the 17 level-1 classes, the OA of the 1975 map decreased to 71.60% (Kappa = 0.68), as expected given the increased classification complexity and the limited spectral information provided by Landsat MSS data. Among all subclasses, mixed forest (MFT) exhibited the lowest accuracy (PA = 48.26%, UA = 42.36%), with most samples being misclassified as evergreen needle-leaved forest (ENF), deciduous needle-leaved forest (DNF), or evergreen broad-leaved forest (EBF). This confusion arises because mixed forests share spectral characteristics with both broad-leaved and needle-leaved forest types, while the four-band MSS data—lacking SWIR information and dense temporal observations—cannot effectively capture subtle differences in canopy composition[52]. Similarly, shrubland (SHR), grassland (GRS), and sparse vegetation (SVG) showed PA values of approximately 55%, primarily because of spectral confusion among herbaceous vegetation, woody vegetation, and bare land in arid and semi-arid transition zones. Inland wetland (IWL) achieved a PA of 62.59% and was mainly confused with water bodies and vegetation. Such misclassification due to spectral confusion among subclasses is unavoidable, and classification accuracy generally decreases as the classification system becomes more refined. Similar patterns have been reported for

	Unchanged	Changed	Total	P.A. (%)	F1 (%)
Unchanged	1088	191	1279	85.07	85.79
Changed	165	220	385	57.14	55.28
Total	1253	411	1664		
U.A. (%)	86.83	53.53			
O.A. (%)	78.61				

Table 4. Error matrix for changed and unchanged samples.

GLC_FCS10[53] and GLC_FCS30[38]. These results further highlight the inherent limitations of early MSS imagery for detailed subclass discrimination. The 1980 dataset showed comparable performance, achieving an OA of 80.08% (Kappa = 0.75) for the 10 basic classes and an OA of 70.71% (Kappa = 0.68) for the 17 level-1 classes.

The error matrix for changed and unchanged samples is presented in Table 4. Given the class imbalance between changed and unchanged samples, the F1-score—the harmonic mean of PA and UA, as defined in Eq. (11)—was adopted as a comprehensive evaluation metric to account for both omission and commission errors. The OA of this binary validation reached 78.61%. Specifically, unchanged points achieved high accuracy, with a PA of 85.07%, a UA of 86.83%, and an F1-score of 85.79%. In contrast, changed points exhibited notably lower performance, with a PA of 57.14%, a UA of 53.53%, and an F1-score of 55.28%, indicating that detecting land cover change is considerably more challenging than identifying stable areas. This observation is further supported by earlier research reporting similar patterns. For instance, the LCMAP_Val dataset-based validation of GLC_FCS30D showed that changed pixels had PA (72.26%) and UA (56.62%) substantially lower than those of unchanged pixels[4].

$$F1 = 2 \times \frac{P.A. \times U.A.}{P.A. + U.A.} \times 100\% \tag{11}$$

Qualitative assessment of representative regions. To demonstrate the capability of GLC_FCS60 to capture early land cover changes, five representative regions were selected across diverse geographic and environmental conditions. Figure 6 presents Landsat MSS false-color composites, land cover maps for 1975, 1980, and 1985, together with the corresponding CCDM outputs and land cover change results for the 1985–1980 and 1985–1975 periods for two representative regions within the Amazon rainforest. Figure 7 presents three additional examples from other typical regions, including Tianjin City, China; the northwestern Eyre Peninsula, South Australia; and DeWitt County, Illinois, United States.

In the first region, the MSS false-color composites reveal progressive fragmentation and lightening of deep-red forest areas over time, indicating ongoing deforestation associated with agricultural expansion. The generated land cover maps clearly capture this transition from forest to cropland. Because CCDM uses 1985 as the reference year, widespread BI signals are detected in this region. These signals correspond to areas where biomass in 1975/1980 was higher than in 1985, reflecting forest-to-cropland conversion and associated biomass loss over time. The detected changes show agreement with the land cover change maps. The second region exhibits more complex land cover transitions, including forest-to-cropland, forest-to-grassland, and forest-to-impervious surface conversions. The imagery shows both a reduction in deep-red forest signatures and the emergence of lighter impervious or cleared areas, consistent with agricultural expansion and urban development. Both the change detection results and the land cover change maps successfully identify these deforestation and land conversion processes.

The three additional examples in Fig. 7 further demonstrate the ability of GLC_FCS60 to capture diverse land cover changes beyond the Amazon region. In Tianjin City, China, cropland and a small proportion of wetlands were converted to impervious surfaces between 1975 and 1985, reflecting progressive urban expansion. In the northwestern Eyre Peninsula, South Australia, forest areas were converted to cropland and shrubland during 1975–1985. In DeWitt County, Illinois, United States, areas of cropland and grassland were converted to water between 1975 and 1980, corresponding to the formation of Clinton Lake during this period. Together, these examples demonstrate that the improved CCDM and locally adaptive classification framework can effectively capture diverse land cover changes while maintaining spatial detail and temporal consistency. The GLC_FCS60 products successfully reproduce major land cover change patterns across different regions from the mid-1970s to the early 1980s, supporting their application in long-term global land cover change studies.

Usage Notes

The GLC_FCS60 dataset provides the first global 60 m land cover maps with a fine classification system for the mid-1970s and early 1980s, filling a critical gap in long-term land cover monitoring. Developed using an improved bi-temporal CCDM, a globally distributed and temporally stable training sample pool, locally adaptive random forest classification, and post-classification optimization, the dataset for 1975 achieved an OA of 80.13% for 10 basic classes and 71.60% for 17 Level-1 classes. Quantitative and qualitative evaluations indicate that the dataset effectively captures major land cover patterns and long-term land cover changes.

Despite these achievements, several limitations should be acknowledged. First, incomplete MSS coverage in the target years (e.g., northeastern Russia in 1980) and persistent cloud contamination in some regions (e.g., Southeast Asia) prevented reliable bi-temporal change detection in some areas. These areas were therefore inherited from GLC_FCS30D_1985. Notably, 21.96% of the 1975 mapping grids were not updated, slightly exceeding the 19.88% without valid MSS observations, because some areas in northeastern Russia had MSS observations in

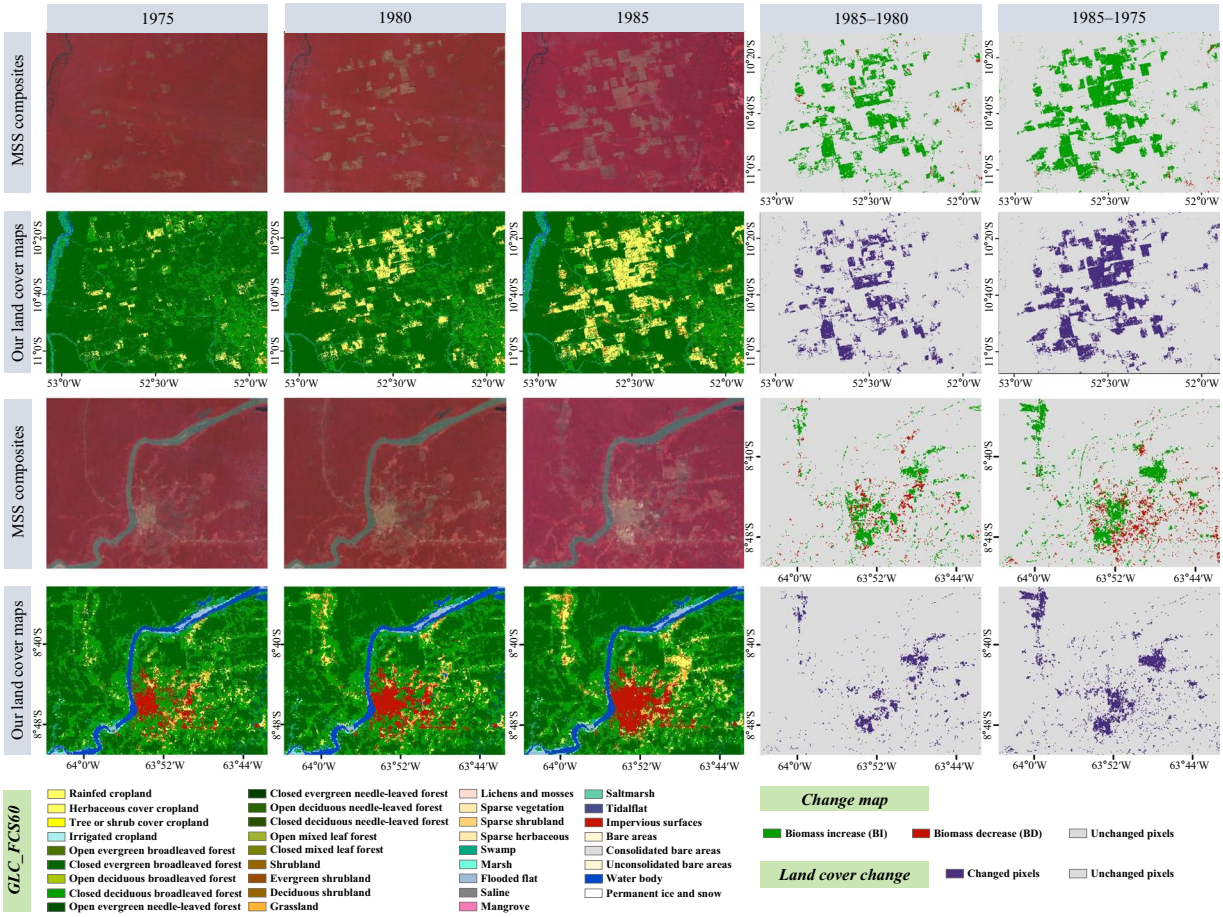


Fig. 6 Examples of land cover changes in two representative regions of the Amazon from 1975 to 1985, including Landsat MSS false-color composites, GLC_FCS60 land cover maps, CCDM change detection results, and derived land cover change maps. The two regions are located at 51.9105–53.0833°W, 10.1839–11.1313°S (Region 1) and 63.6921–64.0602°W, 8.5659–8.8461°S (Region 2).

1975 but lacked valid observations for the 1985 reference period and were therefore also not updated. The spatial distribution of these areas is provided in Figure S1 of the Supplementary Information, and users should consider this limitation when using the data in these areas. Second, resampling GLC_FCS30D from 30 m to 60 m using majority-rule aggregation introduced mixed pixels and reduced spatial detail, causing small rural impervious surface features (e.g., isolated buildings and farmsteads) to be underestimated or absorbed into surrounding classes. Third, relative radiometric normalization could not completely eliminate spectral differences between MSS and TM because of their different spectral response functions[54], potentially affecting change detection accuracy in regions where TM imagery was used as a substitute for MSS observations. Fourth, because MSS lacks SWIR bands, dGNDVI was used to replace the dNBR employed in the original CCDM. However, dGNDVI is generally less sensitive to fire-related disturbances than dNBR, which may reduce change detection performance in fire-prone regions. Fifth, although transitions from impervious to non-impervious surfaces are generally uncommon over the considered period, genuine transitions may occur at the global scale due to building demolition, industrial land abandonment, urban redevelopment, or green space restoration. The irreversibility constraint applied during post-processing may therefore exclude some genuine impervious surface decreases. Users focusing on urban shrinkage, industrial land redevelopment, or fine-scale urban land-use dynamics should be aware of this limitation. Finally, the validation strategy has inherent limitations. For unchanged samples, GLC_FCS30D was used for initial candidate screening—although samples in change areas were fully interpreted and 10% of stable samples were manually checked, some dependency on the reference product is unavoidable. For changed samples, visual interpretation of 60 m MSS imagery is limited for distinguishing detailed subclasses. We mitigated this with a multi-evidence framework, yet ambiguity remains, particularly for the 17 level-1 classes.

Future work will focus on addressing the limitations discussed above. We plan to explore deep-learning-based super-resolution[55] to enhance the spatial resolution of MSS imagery from 60 m to 30 m by leveraging overlapping MSS-TM observations and generating pseudo-TM imagery with a reconstructed SWIR band. This approach has the potential to provide three main advantages: (1) it may reduce information loss associated with resampling 30 m reference data to 60 m, thereby preserving fine spatial details such as small rural impervious surface features; (2) it may improve the consistency of relative radiometric normalization by enabling comparisons at a common 30 m resolution with harmonized spectral bands; and (3) it would provide a reconstructed SWIR band,

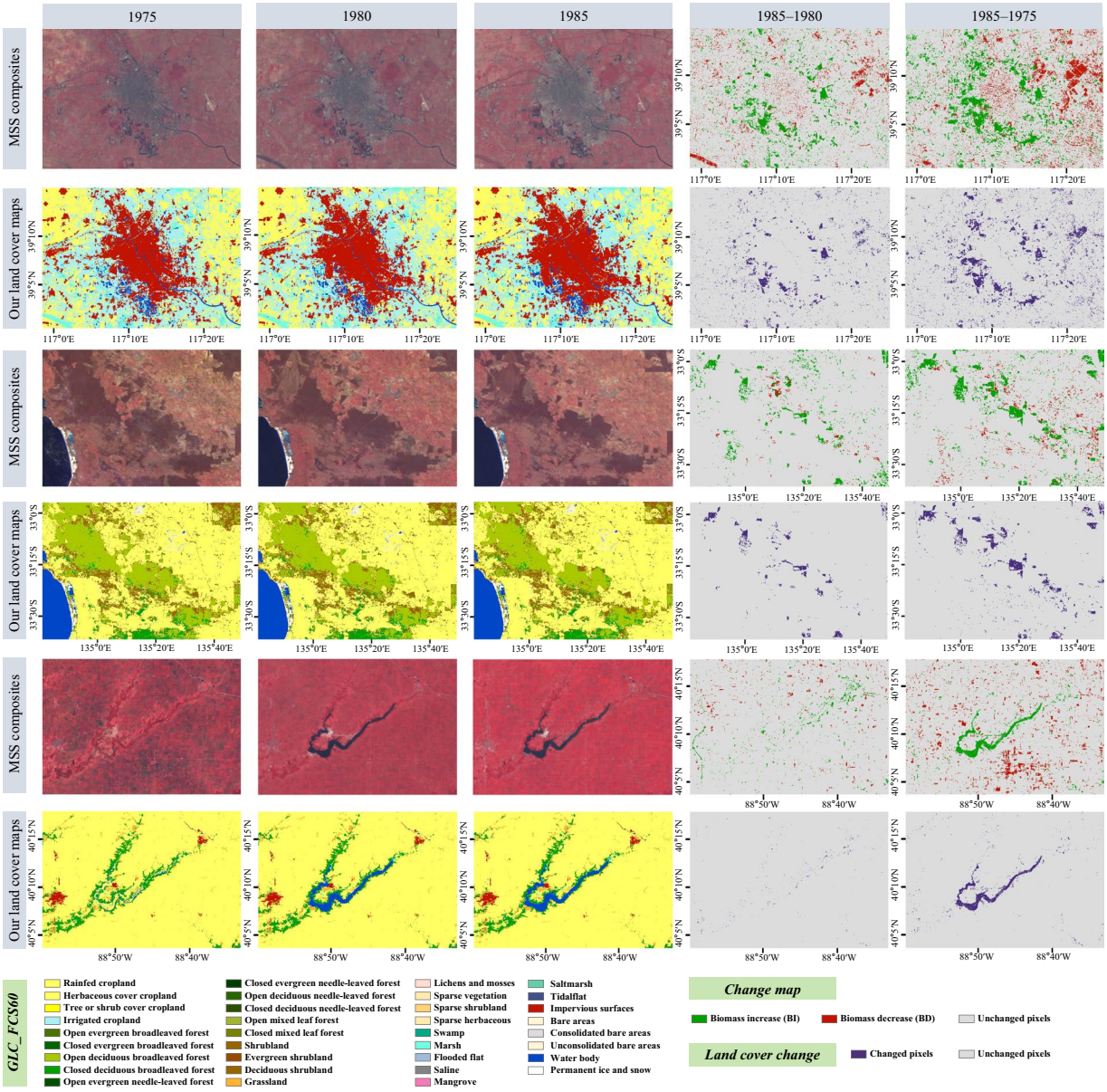


Fig. 7 Examples of land cover changes in three additional global regions from 1975 to 1985, including Landsat MSS false-color composites, GLC_FCS60 land cover maps, improved CCDM change detection results, and derived land cover change maps. The three regions are located in Tianjin, China (116.9782–117.4163°E, 39.0125–39.2469°N), northwestern Eyre Peninsula, South Australia (134.6932–135.8166°E, 32.9463–33.5962°S), and DeWitt County, Illinois, United States (88.5554–88.9918°W, 40.0674–40.2955°N).

thereby enabling the use of dNBR for potentially more sensitive change detection, particularly in fire-prone regions. In addition, we plan to integrate AVHRR time-series data[56] to better characterize phenological variability and reduce observational gaps in persistently cloudy regions. Further efforts to develop more independent and reliable historical reference samples by integrating multi-source historical imagery and ancillary data could improve the reliability of accuracy assessment, particularly for spectrally similar land-cover subclasses.

Data availability

The GLC_FCS60 dataset is freely available via Zenodo at https://doi.org/10.5281/zenodo.20688228[49].

Code availability

The GLC_FCS60 dataset was generated on the GEE platform, with relevant code written in JavaScript and publicly accessible on GitHub (https://github.com/taogerniu/GLC_FCS60-Code/blob/main/GLC_FCS60-Code).

Received: 15 June 2026; Accepted: 21 September 2026;
Published: xx xx xxxx

References

1. Song, X. *et al*. Global land change from 1982 to 2016. *Nature* **560**, 639–643 https://doi.org/10.1038/s41586-018-0411-9 (2018).
2. Luo, Y. *et al*. Hybrid Global Annual 1-km IGBP Land Cover Maps for the Period 2000–2020. *J. Remote Sens.* **4**, 0122 https://doi.org/10.34133/remotesensing.0122 (2024).
3. Gu, Z. & Zeng, M. The Use of Artificial Intelligence and Satellite Remote Sensing in Land Cover Change Detection: Review and Perspectives. *Sustainability* **16**, 274 https://doi.org/10.3390/su16010274 (2024).
4. Zhang, X. *et al*. GLC_FCS30D: the first global 30 m land-cover dynamics monitoring product with a fine classification system for the period from 1985 to 2022 generated using dense-time-series Landsat imagery and the continuous change-detection method. *Earth Syst. Sci. Data* **16**, 1353–1381 https://doi.org/10.5194/essd-16-1353-2024 (2024).
5. Li, B. *et al*. ALCC: Temporally Consistent Annual Land Cover Maps over China from 1985 to 2022 Based on an Ensemble Change Detection Method. *J. Remote Sens.* **6**, 1029 https://doi.org/10.34133/remotesensing.1029 (2026).
6. Gorelick, N. *et al*. Google Earth Engine: Planetary-scale geospatial analysis for everyone. *Remote Sens. Environ.* **202**, 18–27 https://doi.org/10.1016/j.rse.2017.06.031 (2017).
7. Gong, P. *et al*. Finer resolution observation and monitoring of global land cover: first mapping results with Landsat TM and ETM+ Data. *Int. J. Remote Sens.* **34**, 2607–2654 https://doi.org/10.1080/01431161.2012.748992 (2013).
8. Chen, J. *et al*. Global land cover mapping at 30 m resolution: A POK-based operational approach. *ISPRS J. Photogramm. Remote Sens.* **103**, 7–27 https://doi.org/10.1016/j.isprsjprs.2014.09.002 (2015).
9. Liu, H. *et al*. Production of global daily seamless data cubes and quantification of global land cover change from 1985 to 2020-iMap World 1.0. *Remote Sens. Environ.* **258**, 112364 https://doi.org/10.1016/j.rse.2021.112364 (2021).
10. Brown, C. F. *et al*. Dynamic World, Near real-time global 10 m land use land cover mapping. *Sci. Data* **9**, https://doi.org/10.1038/s41597-022-01307-4 (2022).
11. Friedl, M. A. *et al*. Medium Spatial Resolution Mapping of Global Land Cover and Land Cover Change Across Multiple Decades From Landsat. *Front. Remote Sens.* **3**, 894571 https://doi.org/10.3389/frsen.2022.894571 (2022).
12. Yu, L. *et al*. FROM-GLC Plus 3.0: Multimodal Land Change Mapping with SAM and Dense Surface Observations. *J. Remote Sens.* **5**, 0728 https://doi.org/10.34133/remotesensing.0728 (2025).
13. Ai, W. *et al*. Land cover mapping of China based on an improved bi-temporal comprehensive change detection method from Landsat MSS observations. *Int. J. Digit. Earth* **18**, https://doi.org/10.1080/17538947.2025.2597065 (2025).
14. Winkler, K., Fuchs, R., Rounsevell, M. & Herold, M. Global land use changes are four times greater than previously estimated. *Nat. Commun.* **12**, https://doi.org/10.1038/s41467-021-22702-2 (2021).
15. Xu, X. *et al*. China's National Land Use and Cover Change Dataset (CNLUCC). *Resource and Environmental Science Data Registration and Publication System* [Dataset]. https://www.resdc.cn/DOI/DOI.aspx?DOIID=54 (2018).
16. Liu, J. *et al*. Spatiotemporal characteristics, patterns, and causes of land-use changes in China since the late 1980s. *J. Geogr. Sci.* **24**, 195–210 https://doi.org/10.1007/s11442-014-1082-6 (2014).
17. Costa, H., Carrao, H., Bacao, F. & Caetano, M. Combining per-pixel and object-based classifications for mapping land cover over large areas. *Int. J. Remote Sens.* **35**, 738–753 https://doi.org/10.1080/01431161.2013.873151 (2014).
18. Zhu, Z. Change detection using landsat time series: A review of frequencies, preprocessing, algorithms, and applications. *ISPRS J. Photogramm. Remote Sens.* **130**, 370–384 https://doi.org/10.1016/j.isprsjprs.2017.06.013 (2017).
19. Liu, L. *et al*. Finer-Resolution Mapping of Global Land Cover: Recent Developments, Consistency Analysis, and Prospects. *J. Remote Sens.* **2021**, 5289697 https://doi.org/10.34133/2021/5289697 (2021).
20. Xian, G., Homer, C. & Fry, J. Updating the 2001 National Land Cover Database land cover classification to 2006 by using Landsat imagery change detection methods. *Remote Sens. Environ.* **113**, 1133–1147 https://doi.org/10.1016/j.rse.2009.02.004 (2009).
21. Yan, J. *et al*. A time-series classification approach based on change detection for rapid land cover mapping. *ISPRS J. Photogramm. Remote Sens.* **158**, 249–262 https://doi.org/10.1016/j.isprsjprs.2019.10.003 (2019).
22. Zhu, Z. & Woodcock, C. E. Continuous change detection and classification of land cover using all available Landsat data. *Remote Sens. Environ.* **144**, 152–171 https://doi.org/10.1016/j.rse.2014.01.011 (2014).
23. Jin, S. *et al*. A comprehensive change detection method for updating the National Land Cover Database to circa 2011. *Remote Sens. Environ.* **132**, 159–175 https://doi.org/10.1016/j.rse.2013.01.012 (2013).
24. Li, Z. *et al*. Post-Processing Optimization of the Global 30 m Land Cover Dynamic Monitoring Product. *Remote Sens* **17**, 1558 https://doi.org/10.3390/rs17091558 (2025).
25. Liu, L., Zhang, X. & Li, Z. GLC_FCS30D: the first global 30-m land-cover dynamic monitoring product with fine classification system from 1985 to 2022. *Zenodo* https://doi.org/10.5281/zenodo.15063683 (2025).
26. NASA/METI/AIST/Japan Spacesystems and U.S./Japan ASTER Science Team. ASTER Global Digital Elevation Model V003 [Dataset]. https://doi.org/10.5067/ASTER/ASTGTM.003 (2019).
27. Zhao, T. *et al*. Assessing the Accuracy and Consistency of Six Fine-Resolution Global Land Cover Products Using a Novel Stratified Random Sampling Validation Dataset. *Remote Sens* **15**, 2285 https://doi.org/10.3390/rs15092285 (2023).
28. Liu, L., Zhao, T. & Zhang, X. A novel stratified random sampling global validation dataset in 2020—SRS_Val dataset. *Zenodo* https://doi.org/10.5281/zenodo.7846090 (2023).
29. Jin, S. *et al*. National Land Cover Database 2019: A New Strategy for Creating Clean Leaf-On and Leaf-Off Landsat Composite Images. *J. Remote Sens.* **3**, 0022 https://doi.org/10.34133/remotesensing.0022 (2023).
30. Jin, S. *et al*. National Land Cover Database 2019: A comprehensive Strategy for Creating the 1986-2019 Forest Disturbance Product. *J. Remote Sens.* **3**, 0021 https://doi.org/10.34133/remotesensing.0021 (2023).
31. Nielsen, A. A. The regularized iteratively reweighted MAD method for change detection in multi- and hyperspectral data. *IEEE Trans. Image Process.* **16**, 463–478 https://doi.org/10.1109/tip.2006.888195 (2007).
32. Canty, M. J., Nielsen, A. A. & Schmidt, M. Automatic radiometric normalization of multitemporal satellite imagery. *Remote Sens. Environ.* **91**, 441–451 https://doi.org/10.1016/j.rse.2003.10.024 (2004).
33. Taddeo, S., Dronova, I. & Depsky, N. Spectral vegetation indices of wetland greenness: Responses to vegetation structure, composition, and spatial distribution. *Remote Sens. Environ.* **234**, 111467 https://doi.org/10.1016/j.rse.2019.111467 (2019).
34. Mangewa, L. J. *et al*. Comparative assessment of UAV and sentinel-2 NDVI and GNDVI for preliminary diagnosis of habitat conditions in Burunge wildlife management area, Tanzania. *Earth* **3**, 769–787 https://doi.org/10.3390/earth3030044 (2022).
35. Wang, Y. *et al*. Automatic training sample collection utilizing multi-source land cover products and time-series Sentinel-2 images. *GISci. Remote Sens.* **61**, 2352957 https://doi.org/10.1080/15481603.2024.2352957 (2024).
36. Zhang, X. *et al*. Development of a global 30m impervious surface map using multisource and multitemporal remote sensing datasets with the Google Earth Engine platform. *Earth Syst. Sci. Data* **12**, 1625–1648 https://doi.org/10.5194/essd-12-1625-2020 (2020).
37. Xiao, Y., Wang, Q. & Zhang, H. K. Global Natural and Planted Forests Mapping at Fine Spatial Resolution of 30 m. *J. Remote Sens.* **4**, 0204 https://doi.org/10.34133/remotesensing.0204 (2024).
38. Zhang, X. *et al*. GLC_FCS30: global land-cover product with fine classification system at 30m using time-series Landsat imagery. *Earth Syst. Sci. Data* **13**, 2753–2776 https://doi.org/10.5194/essd-13-2753-2021 (2021).
39. Zhang, X. *et al*. GWL_FCS30: a global 30 m wetland map with a fine classificationsystem using multi-sourced and time-series remote sensing imagery in 2020. *Earth Syst. Sci. Data* **15**, 265–293 https://doi.org/10.5194/essd-15-265-2023 (2023).
40. Jin, H., Stehman, S. V. & Mountrakis, G. Assessing the impact of training sample selection on accuracy of an urban classification: a case study in Denver, Colorado. *Int. J. Remote Sens.* **35**, 2067–2081 https://doi.org/10.1080/01431161.2014.885152 (2014).

41. Millard, K. & Richardson, M. On the Importance of Training Data Sample Selection in Random Forest Image Classification: A Case Study in Peatland Ecosystem Mapping. *Remote Sens* **7**, 8489–8515 https://doi.org/10.3390/rs70708489 (2015).
42. Zhu, Z. *et al*. Optimizing selection of training and auxiliary data for operational land cover classification for the LCMAP initiative. *ISPRS J. Photogramm. Remote Sens.* **122**, 206–221 https://doi.org/10.1016/j.isprsjprs.2016.11.004 (2016).
43. Zhang, X. *et al*. GISD30: global 30 m impervious-surface dynamic dataset from 1985 to 2020 using time-series Landsat imagery on the Google Earth Engine platform. *Earth Syst. Sci. Data* **14**, 1831–1856 https://doi.org/10.5194/essd-14-1831-2022 (2022).
44. Mellor, A., Boukir, S., Haywood, A. & Jones, S. Exploring issues of training data imbalance and mislabelling on random forest performance for large area land cover classification using the ensemble margin. *ISPRS J. Photogramm. Remote Sens.* **105**, 155–168 https://doi.org/10.1016/j.isprsjprs.2015.03.014 (2015).
45. Gong, P. *et al*. Stable classification with limited sample: transferring a 30-m resolution sample set collected in 2015 to mapping 10-m resolution global land cover in 2017. *Sci. Bull.* **64**, 370–373 https://doi.org/10.1016/j.scib.2019.03.002 (2019).
46. Zhang, X. *et al*. Global annual wetland dataset at 30 m with a fine classification system from 2000 to 2022. *Sci. Data* **11**, 310 https://doi.org/10.1038/s41597-024-03143-0 (2024).
47. Qiu, L., He, J., Yue, C., Ciais, P. & Zheng, C. Substantial terrestrial carbon emissions from global expansion of impervious surface area. *Nat. Commun.* **15**, 6456 https://doi.org/10.1038/s41467-024-50840-w (2024).
48. Jin, S., Yang, L., Zhu, Z. & Homer, C. A land cover change detection and classification protocol for updating Alaska NLCD 2001 to 2011. *Remote Sens. Environ.* **195**, 44–55 https://doi.org/10.1016/j.rse.2017.04.021 (2017).
49. Liu, L., Zhang, X. & Ai, W. GLC_FCS60: the first global 60-m land cover product with a fine classification system for 1975 and 1980. *Zenodo* https://doi.org/10.5281/zenodo.20688228 (2026).
50. Zhao, T., Zhang, X. & Liu, L. An interactive toolkit based on Google Earth Engine for visual interpretation and analysis of land cover and land cover changes. *Big Earth Data*, 1-25, https://doi.org/10.1080/20964471.2025.2608464 (2025).
51. Olofsson, P. *et al*. Good practices for estimating area and assessing accuracy of land change. *Remote Sens. Environ.* **148**, 42–57 https://doi.org/10.1016/j.rse.2014.02.015 (2014).
52. Buitrago, M. F., Skidmore, A. K., Groen, T. A. & Hecker, C. A. Connecting infrared spectra with plant traits to identify species. *ISPRS J. Photogramm. Remote Sens.* **139**, 183–200 https://doi.org/10.1016/j.isprsjprs.2018.03.013 (2018).
53. Zhang, X. *et al*. GLC_FCS10: a global 10 m land-cover dataset with a fine classification system from Sentinel-1 and Sentinel-2 time-series data in Google Earth Engine. *Earth Syst. Sci. Data* **17**, 4039–4062 https://doi.org/10.5194/essd-17-4039-2025 (2025).
54. Roy, D. P. *et al*. Characterization of Landsat-7 to Landsat-8 reflective wavelength and normalized difference vegetation index continuity. *Remote Sens. Environ.* **185**, 57–70 https://doi.org/10.1016/j.rse.2015.12.024 (2016).
55. Xiong, S. *et al*. Retrieving Historical Images at 10-m Resolution From 1985 to 2015 Through STARS-Net: A Spatiotemporal Attention Referenced Super-Resolution Network. *IEEE Trans. Geosci. Remote Sens.* **63**, 5401119 https://doi.org/10.1109/tgrs.2025.3530464 (2025).
56. Nelson, R. Regression and ratio estimators to integrate AVHRR and MSS data. *Remote Sens. Environ.* **30**, 201–216 https://doi.org/10.1016/0034-4257(89)90062-X (1989).

Author contributions

W.A. designed and implemented this research, produced the dataset, and wrote the manuscript. X.Z. supervised the research design and methodology, and revised the manuscript. C.W.and F.T. implemented data processing and collected validation datasets. T.Z. and W.Z. contributed to accuracy validation. L.L. provided overall methodological guidance, and reviewed the manuscript.

Funding

This research was supported by the National Key Research and Development Program of China [grant number 2023YFB3907403].

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41597-026-08368-9.

Correspondence and requests for materials should be addressed to X.Z.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

© The Author(s) 2026

This Article in Press is shared early to give you faster access to new research. It is citable and carries a permanent DOI. The final edited version will replace it automatically.